Standard Series

Questions 1–46

Question 1. Find:

a) $\sum_{r=1}^{30} 8$

b) $\sum_{r=1}^{50} r$

c) $\sum_{r=1}^{20} r^2$

d) $\sum_{r=1}^{10} r^3$

Question 2. Find an expression for each of the following using standard series formulae. Fully expand your answer:

a) $\sum_{r=1}^{n} (2r+3)$

b) $\sum_{r=1}^{n} (5r-12)$

c) $\sum_{r=1}^{n} (4r+1)$

d) $\sum_{r=1}^{n} (ar+b)$
where a and b are constants.

Question 3. Find the positive integer k such that

$$\sum_{r=1}^{k}(2r+k)=300$$

Question 4. Using the standard results for summations, show that

$$\sum_{r=1}^{12}r(6r-1)=3822$$

Question 5. Using the standard results for summations, find the value of

$$\sum_{r=1}^{50}(3r+2)^2$$

Question 6. Fully factorise each of the following:

a) $n(n+1)+4n$
b) $\frac{7}{2}n(n+1)-5n$
c) $3n(n+1)(2n+1)+\frac{1}{2}n(n+1)$
d) $4n(n+1)(2n+1)-\frac{1}{4}n^2(n+1)^2$
e) $\frac{1}{2}n^2(n+1)^2-3n(n+1)$
f) $\frac{1}{4}n^2(n+1)^2-\frac{1}{6}n(n+1)(2n+1)$
g) $\frac{3}{4}n^2(n+1)^2+\frac{5}{2}n(n+1)$

Question 7. Find a fully factorised expression for each of the following using standard series formulae:

a) $\sum_{r=1}^{n}(r^3+r^2)$

b) $\sum_{r=1}^{n}(2r^2+5r)$

c) $\sum_{r=1}^{n}(7r^3-2r^2)$

d) $\sum_{r=1}^{n}(1-2r+5r^3)$

e) $\sum_{r=1}^{n}r(2r+1)$

f) $\sum_{r=1}^{n}r^2(r-4)$

Question 8. Using the standard results for summations, show that

$$\sum_{r=1}^{n}(4r-2)^2=\frac{4}{3}n(2n-1)(2n+1)$$

Question 9. Using the standard results for summations, find the sum of the first n terms of each of the following series:

a) $(1 \times 2) + (2 \times 3) + (3 \times 4) + \ldots$
b) $(3 \times 5) + (4 \times 6) + (5 \times 7) + \ldots$
c) $(2 \times 4) + (3 \times 5) + (4 \times 6) + \ldots$
d) $2 \times 1^2 + 3 \times 2^2 + 4 \times 3^2 + \ldots$
e) $(1 \times 3 \times 4) + (2 \times 4 \times 5) + (3 \times 5 \times 6) + \ldots$

Question 10.

a) Using the standard results for summations, find an expression for

$$\sum_{r=1}^{n}(r+1)(3r-1)$$

b) Hence evaluate

$$2 \times 2 + 3 \times 5 + 4 \times 8 + \ldots + 51 \times 149$$

Question 11. Find a fully factorised expression for

a) $\sum_{r=1}^{2n} r^2$

b) $\sum_{r=1}^{2n} (2r)^2$

c) $1^2 + 3^2 + 5^2 + \ldots + (4n-1)^2$

Question 12. Using the standard results for summations, show that

a) the sum of the square of the first 10 consecutive even numbers is 1540

b) the sum of the square of the first 10 consecutive odd numbers is 1330

c) the sum of the cubes of the first 10 consecutive even numbers is 24 200

Question 13. Find the sum of all the integers from 1 to 999 inclusive that are not square numbers.

Question 14. Given that the sum of the squares of the first n consecutive even numbers is 82 160, find the value of n

Question 15. Given that the sum of the squares of the first n consecutive odd numbers is 20 825, find the value of n

Question 16.

a) By considering the standard results for summations, find

$$24 + 3\sum_{r=1}^{n}(r+2)(r+3)$$

as a product of three linear factors.

b) Hence, show that

$$24 + 3\sum_{r=1}^{n}(r+2)(r+3)$$

is divisible by 6

Question 17.

a) Show that

$$\sum_{r=1}^{n}(7r-2)^2 = \frac{1}{6}n(an^2+bn+c)$$

where a, b and c are constants to be found.

b) Hence find the value of k for which

$$\sum_{r=1}^{k}(7r-2)^2 = 190k^2$$

Question 18. Given that $\mathrm{f}(r) = ar + b$ for constants a and b,

$$\sum_{r=1}^{3}\mathrm{f}(r) = 0$$

and

$$\sum_{r=1}^{7}\mathrm{f}(r) = 35$$

a) Find a general expression for

$$\sum_{r=1}^{n}\mathrm{f}(r)$$

b) Hence find

$$\sum_{r=1}^{20}\mathrm{f}(r)$$

Question 19.

a) Show that

$$\sum_{r=1}^{n}(5r-7)=\frac{1}{2}n(5n-9)$$

b) Find the smallest value of k for which

$$\sum_{r=1}^{k}(5r-7)>6000$$

Question 20. Find constants a and b such that

$$\sum_{r=1}^{n}(ar+b)=n^2-10n$$

Question 21. An art gallery has a collection of n paintings numbered from 1 to n, which are rectangular with lengths $(r + 10)$ cm and $(r + 12)$ cm, where r is the number of each painting.

a) Show that the formula for the total area of n paintings is $\frac{1}{6}n(2n^2 + 69n + 787)$ cm^2.

b) Given that the total area of the n paintings is 3806 cm^2, calculate how many paintings there are in the collection.

Question 22. A company manufacturing bicycles agrees to a 15-year contract with a supplier to provide bicycles annually. In the first year of the contract, 250 bicycles are delivered and each subsequent year the delivery increases by 40.

a) State an expression for the number of bicycles supplied in year r

In year r the price per bicycle (£) is given by $50 + \frac{r^2}{30}$ and the total cost of producing all the bicycles delivered that year (in £) is given by

$$2000 + 7r^2$$

b) Show that the profit made by the company in year r is

$$\frac{4}{3}(r^3 + 1500r + 6375)$$

c) Using the standard results for summations, show that the total profit made by the company in the first n years is

$$\frac{1}{3}n(n^3 + an^2 + bn + c)$$

where a, b and c are constants to be found.

After 15 years the actual profit of the company is £280 000

d) Comment on the validity of the model.

Prerequisite knowledge for Q23: Standard Deviation

Question 23. A bakery is preparing a special display of n loaves of bread for the shop window. The loaves are arranged in ascending order of weight. The first loaf weighs 400 grams and each subsequent loaf is 3 grams heavier.

a) State the mass of the rth loaf in terms of r

b) Show that the average weight, in grams, of the n loaves is given by

$$\frac{1}{2}(3n + 797)$$

There are 20 loaves in the display.

c) Using the standard results for summation, calculate the standard deviation, given by

$$\sqrt{\frac{\sum r^2}{n} - \left(\frac{\sum r}{n}\right)^2}$$

of the weights of the loaves in the display, giving your answer to 3 significant figures.

Question 24.

a) Show that

$$\frac{1^3 + 2^3 + 3^3 + \ldots + n^3}{1 + 2 + 3 + \ldots + n} = \frac{1}{2}n(n + 1)$$

b) Hence, find the smallest value of n such that

$$\frac{1^3 + 2^3 + 3^3 + \ldots + n^3}{1 + 2 + 3 + \ldots + n} > 200$$

Question 25.

a) Find a fully factorised expression for

$$\sum_{r=1}^{n}(r^3 - 2r)$$

b) Find the value of k such that

$$\sum_{r=1}^{k}(2r^3 + 3r) - \sum_{r=1}^{k}(r^3 + 5r) = 1224$$

Summation with Variable Lower Bounds

Question 26. Rewrite each of the following in sigma notation starting at $r = 1$:

a) $\sum_{r=10}^{30}(r-15)(r-16)$

b) $\sum_{r=4}^{16}(r^2+3r)$

c) $\sum_{r=12}^{20}(3r^3-1)$

d) $\sum_{r=1}^{8}r^2 - \sum_{r=1}^{4}r^2$

Question 27. Using the standard summation results, find:

a) $\sum_{r=10}^{20}(r^2-3)$

b) $\sum_{r=5}^{10} r(r-1)$

c) $\sum_{r=50}^{100}(r^3-1)$

Question 28. Using the standard results for summations, evaluate $51^2+52^2+53^2+\ldots+101^2$

Question 29.

a) Using the standard results for summations, show that

$$\sum_{r=1}^{n} r(r-1)(r+1) = \frac{1}{4}n(n-1)(n+1)(n+2)$$

b) Hence, show that

$$\sum_{r=1}^{2n} r(r-1)(r+1) = n(n+1)(2n-1)(2n+1)$$

Question 30. Find fully factorised expressions for each of the following using standard series formulae:

a) $\sum_{r=1}^{2n} r^3$

b) $\sum_{r=n+1}^{2n} r^2$

c) $\sum_{r=1}^{n+4} r^2$

d) $\sum_{r=1}^{n+3} (r^3 - 2r^2)$

e) $\sum_{r=n+1}^{2n} (r^3 + r)$

Question 31. Show that, for $n \geq 6$

$$\sum_{r=6}^{n+1} r^2 = \frac{1}{6}(an^3 + bn^2 + cn + d)$$

where a, b, c and d are constants to be found.

Question 32.

a) Show that, for values of a and b to be found that

$$\sum_{r=1}^{n} r(r-5) = \frac{1}{3}n(n-a)(n+b)$$

b) Hence, find a fully factorised expression for

$$\sum_{r=n+1}^{3n} r(r-5)$$

Question 33.

a) Find the sum of the squares of the first n positive even numbers.

b) Given that the sum of the squares of 5 consecutive even numbers is 1020, by using standard results for summation, determine the smallest of these consecutive even numbers.

Question 34. Express

$$\sum_{r=1}^{2n} r^3 - \sum_{r=1}^{n} (r+2)^2$$

as a product of three factors in terms of n

Question 35. Given a and b are constant integers, find the possible values of a and b such that

$$\sum_{r=1}^{n}(r+a)(r+b) = \frac{1}{3}n(n^2-7)$$

Question 36.

a) Show that

$$\sum_{r=1}^{2n}(n-r)^2 = \frac{1}{3}n(2n^2+1)$$

b) Use standard summation results to prove that

$$\sum_{r=n+1}^{2n}(n-r)^2 = \sum_{r=1}^{n}r^2$$

Question 37.

a) Show that for positive constants a, b, c and d

$$\sum_{r=1}^{3n-2}(5r+3) = \frac{1}{2}(an-b)(cn+d)$$

b) Hence calculate

$$\sum_{r=1}^{16}(5r+3)$$

Question 38.

a) Show that

$$\sum_{r=1}^{2n+7}(r^2-1) = \frac{1}{3}(4n+19)(2n+7)(n+3)$$

b) Hence calculate

$$\sum_{r=10}^{63}(r^2-1)$$

Series with Alternating and Periodic Terms

Question 39. Find:

a) $\sum_{r=1}^{8}(-1)^r$

b) $\sum_{r=1}^{2n}(-1)^r$

c) $\sum_{r=1}^{9}(-1)^r$

d) $\sum_{r=1}^{2n-1}(-1)^r$

Question 40. Using the standard results for summations, show that:

a) $\sum_{r=1}^{2n}(-1)^r r = n$

b) $\sum_{r=1}^{2n-1}(-1)^r r = -n$

c) $\sum_{r=1}^{2n}(-1)^r(r+1) = n$

d) $\sum_{r=1}^{2n}(-1)^r r^2 = n(2n+1)$

Q41-Q44 use radians

Question 41. Find, giving your answers in exact form:

a) $\sum_{r=1}^{10} \sin\left(\frac{\pi}{2}r\right)$

b) $\sum_{r=1}^{32} r\cos\left(\frac{\pi}{2}r\right)$

c) $\sum_{r=1}^{27} r\tan\left(\frac{\pi}{3}r\right)$

Question 42. Find:

a) $\sum_{r=1}^{n} \sin(\pi r)$

b) $\sum_{r=1}^{n} \cos(2\pi r)$

c) $\sum_{r=1}^{3n} \tan\left(\frac{\pi}{3}r\right)$

Question 43.

a) Show that

$$\sum_{r=1}^{n}(2r+1)^2 = \frac{1}{3}n(4n^2+12n+11)$$

b) Hence find the values of k for which

$$\sum_{r=1}^{k}(2r+1)^2 + 15\sum_{r=1}^{k}\cos(2\pi r) = 24k^2$$

Question 44.

a) Show that

$$\sum_{r=1}^{n}(5r-3)^2 = \frac{1}{6}n(50n^2-15n-11)$$

b) Hence find the value of k for which

$$\sum_{r=3}^{k}(5r-3)^2 + 20\sum_{r=1}^{20} r\sin\left(\frac{\pi}{2}r\right) = 241k$$

Question 45.

a) Using the standard results for summations, show that

$$\sum_{r=1}^{2n} r((-1)^r + 5)^2 = n(52n + 36)$$

b) Hence find

$$\sum_{r=15}^{80} r((-1)^r + 5)^2$$

Question 46.

a) Using the standard results for summations, show that

$$\sum_{r=1}^{2n} ((-1)^r + 3r)^2 = 24n^3 + 18n^2 + 11n$$

b) Hence find

$$\sum_{r=49}^{100} ((-1)^r + 3r)^2$$